

Will Standard-of-Living Indicators Capture the Welfare Effects of Climate Transition Policies?*

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Abstract

Climate policies are designed to reduce future well-being losses from climate change, but they may impose short and medium-run costs on households. This article asks whether standard measures of living standards will accurately record these costs as the transition unfolds. We use a stylized model to compare the evolution of standard indicators with consumer utility under three transition mechanisms: green creative destruction, taxation of carbon-intensive goods, and quantitative restrictions on their consumption. Price-mediated mechanisms are, in principle, largely captured by existing indicators, although new-good effects and non-homothetic preferences generate familiar index-number problems. However, regulatory constraints can reduce welfare without changing either nominal income or observed prices, and can therefore leave measured real income unchanged. We then discuss how equivalent-income methods can be used to monetize such constraints and to incorporate non-monetary co-benefits such as health or environmental quality. Finally, we discuss how the welfare interpretation of chain-weighted real income can partly survive endogenous greening of preferences, while becoming more path-dependent and normatively more fragile.

JEL: C43, E01, E31, I31.

Keywords: Climate transition, Living standards, Well-being, Equivalent income, Cost-of-living indices

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1 Introduction

The treatment of environmental issues is often criticized as a weak point in monetary approaches to living conditions (Gadrey and Jany-Catrice 2016; Laurent and Le Cacheux 2016), whether in the highly aggregated framework of national accounts or in more microeconomic approaches that also examine the distribution of living standards across households. In the case of national accounts, recurring calls have been made to enrich them with sustainability indicators so as to assess, beyond GDP, whether sufficient efforts are being made to ensure that future well-being levels are at least equivalent to current ones. This is a complex undertaking, because the subject is simultaneously prospective, multidimensional and global: prospective, because it involves assessing the impact of current decisions and actions on future living conditions; multidimensional, because these actions and decisions cover many areas - greenhouse gas emissions and climate action are currently prominent, but sustainability is much broader; and global, since the sustainability of living conditions in a given country depends on the actions and decisions of all countries and therefore cannot be measured by accounts for each country in isolation. This explains the slow progress in this area, despite some advances (Germain and Lellouch 2020).

However, there is one aspect of the climate transition that the current statistical system should be better able to cover: measuring its local, real-time costs as it progresses. This topic has long been neglected because of the prevailing belief that these costs could be kept to a minimum, a view that is increasingly being questioned (Pisani-Ferry and Mahfouz 2023). While the climate transition aims to increase long-term well-being relative to a reference scenario without climate policy, short-term costs are expected because it entails phasing out highly energy-intensive production and consumption practices that have driven past growth. When measuring these costs, national accounts and income and price statistics should fulfill their usual role. They have been used to quantify how reliance on fossil fuels and poor environmental practices has facilitated the rise in living standards. They should likewise be capable of measuring the impact that the consumption restrictions required by greening would have on living standards.

But are these measures guaranteed to be comprehensive? Conversely, might we overlook elements that could offset these costs if the transition brings non-monetary co-benefits with sufficiently rapid local effects, without waiting for the expected long-term global benefits? The transition could also be accompanied by a greening of preferences, which should be taken into account since it would reduce the well-being impact of lower consumption of carbon-intensive goods.

This article does not claim to be exhaustive; rather, it explores several of these issues. None is entirely specific to the climate transition, but the transition provides an opportunity to examine them in greater depth or from a new perspective. The article begins by analyzing how the transition could affect households' monetary living standards, a question also addressed by Dees et al. (2023). It focuses on three vectors of decarbonization: a green version of the classic process of creative destruction, taxation on carbon-intensive goods, and regulations restricting their consumption. The current method for calculating purchasing power should, in theory, account for the first two types of greening, but not necessarily for the third. This limitation arises because measurement instruments based on income and prices fail to take account of other factors that can limit consumption opportunities for given levels of income

and prices.

As a direct extension of this first observation, further questions arise concerning indicators related to other aspects of real income, such as the volume/price decomposition of public services from which households benefit, given that these services will also need to be decarbonized (The Shift Project 2023), and the devaluation of carbon-intensive assets held by households. These issues are not explored further here; we refer interested readers to the report on which this article is based (Blanchet, Pesme, and Pommeret 2023). The third section therefore turns directly to the non-monetary co-benefits of the transition. Taking account of non-monetary elements of well-being is a classic problem, and we review the available options for addressing it. The greening of preferences has received much less attention and raises particularly challenging conceptual problems. We examine this issue in the fourth section.

2 Green Transition and Purchasing Power of Disposable Income

The first step is to determine which usual statistical indicators are best suited to capturing the net costs of the climate transition. It is common to focus on the impact on GDP: can greening be compatible with continued GDP growth or will it necessarily lead to a significant slowdown or even a reversal? This first section will instead focus on another indicator from national accounts, namely household gross disposable income (GDI), and on its counterpart in social statistics, the household standard of living.

Broadly speaking, these two indicators represent households' total primary income, which consists mostly of labor income and, for some households, capital income. All taxes and contributions paid by households are deducted from this income, while monetary benefits received are added. The resulting figure is then deflated by a chain-weighted price index: price variations for different goods and services are weighted according to their shares in the household budget, with these shares updated over time. Finally, adjusting real income for household size and for the economies of scale that household size entails gives a measure of the (equivalized) purchasing power of income at the household level. Throughout the article, we use the umbrella terms real income and living standards interchangeably to refer to this concept of purchasing power of income.

From both macro and micro perspectives, this article compares these indicators with a stylized theoretical representation of consumer utility. Such a comparison cannot be dismissed on the grounds that GDI or living standards are not intended to measure well-being. This defensive argument is often used to deflect criticism of these indicators. It is true that none of them is intended to provide a comprehensive measure of well-being. However, they are supposed to capture a key component of well-being: the utility derived from income and the consumption it enables. It might be argued, in response, that this utility cannot be cardinalized unequivocally and that the comparison is therefore meaningless, but this counterargument is also invalid. Although standard-of-living indicators cannot be expected to correspond directly to a cardinal measure of utility, their messages should still be as consistent as possible with ordinal preferences. There would indeed be a significant problem if measures of standard of living suggested an improvement between two periods t and t' while, all else being equal, households in period t' would prefer to return to their nominal income and price

levels of period t . Such a risk can never be completely eliminated, but it is important to ensure that the green transition does not exacerbate it.

2.1 Growth and Renewal of Goods: The Standard Case of Creative Destruction

To understand the risk of contradictory messages, it is useful to first revisit why it is usually ignored. If this risk does not immediately come to mind, it is because growth is often imagined as a scenario in which the consumption of goods and services increases in all dimensions, and having more of all these goods and services is assumed to be inherently preferable.

In reality, growth is never entirely of this kind, because it always goes hand in hand with the renewal of goods: new goods spread and their consumption grows as they replace goods whose consumption declines until they disappear completely. Growth is therefore a process of both addition and subtraction. Most often, however, it has taken the form of a spontaneous process of creative destruction driven by falling prices for new goods, rather than by rising prices for existing goods. In such cases, it is reasonable to assume that consumers benefit from the process and that the additions outweigh the subtractions. The improvement in living standards may even be underestimated because accurately measuring the contribution of goods when they first enter the market is difficult - this is the well-known ‘new goods’ bias. Only once they are fully integrated into the price index can statisticians measure how their falling prices improve living standards.

This familiar line of reasoning is illustrated by a first simulation of the three-good model used throughout this article. It considers a generic good 0 and two goods, 1 and 2, between which growth generates a substitution effect (Box 1 and Figure 1). The simulation starts from an initial situation in which good 2 may exist but can be marketed only at a price unacceptable to the consumer, i.e. above the reservation price beyond which demand for the good is zero. We then assume that a technological shock causes its price to fall significantly below the reservation price at a given time $t_1 = 25$,¹ leading to an immediate jump in both production and demand. After this initial surge, production continues to benefit from technical progress, resulting in further price reductions and increased consumption until time $t_2 = 100$.

— Box 1 - A Three-Good Model

Throughout the article, our discussion is based on a model consisting of three goods: two goods between which substitution will take place, with prices p_1 and p_2 and consumed in quantities q_1 and q_2 , and an aggregate of all other goods consumed in quantity q_0 . For the greening scenarios, good 2 is the green good and good 1 the polluting (“brown”) good. Preferences are represented by a nested CES (Constant Elasticity of Substitution) function, which is maximized under the budget constraint $R = q_0 + p_1 q_1 + p_2 q_2$ and, in some simulations, a regulatory cap $\bar{q}_1$ on consumption of good 1. Goods 1 and 2 are combined in a first CES aggregator, while a second CES aggregator combines them with good q_0 . Incompressible

¹This date could also be interpreted as the date at which good 2 is introduced into the basket of consumer goods used to calculate the price index.

minimum consumption or usage terms B_0 and B are added to this second CES. They can also take negative values, in which case good 0 or the composite good is non-essential, i.e. need not be consumed at all. The overall utility function $U(q_0, q_1, q_2)$ is expressed as follows:

$$U(q_0, q_1, q_2) = \left[a_0(q_0 - B_0)^{\frac{\sigma_0-1}{\sigma_0}} + (1 - a_0) \left(\left(a q_1^{\frac{\sigma-1}{\sigma}} + (1 - a) q_2^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}} - B \right)^{\frac{\sigma_0-1}{\sigma_0}} \right]^{\frac{\sigma_0}{\sigma_0-1}}. \quad (1)$$

In all simulations except the final one, in which preferences vary, the parameter values for this function are fixed at $a_0 = 0.25$, $a = 0.55$, $B_0 = 1$, $B = -1$, $\sigma_0 = 0.5$ and $\sigma = 2$.

What relationship should we expect between this representation of utility and standard-of-living indicators? In general, the relationship can be exact only in the special case of a homogeneous function U of degree 1. In this case, it is possible to write $U = \sum_i U'_i q_i$, from which we can derive the form $U = \lambda \sum_i p_i q_i$ and $dU/U = \sum_i p_i dq_i / \sum_i p_i q_i$, which reflects the variation in volume at current prices. In this case, chain-linking these variations would reproduce the change in U between any two dates.

When the utility function can be expressed as $V = g(U(q))$, where g is any monotone function and U retains the same homogeneity of degree 1, this scalar equivalence is lost, but consistency remains between the measurement of living standards and ordinal preferences: in this case, an increase in income deflated by a chain-weighted price index still corresponds, all else being equal, to an increase in both U and V . This is the case of “homothetic” preferences, in which an equal increase in all quantities consumed has the same effect on well-being regardless of the initial consumption structure.

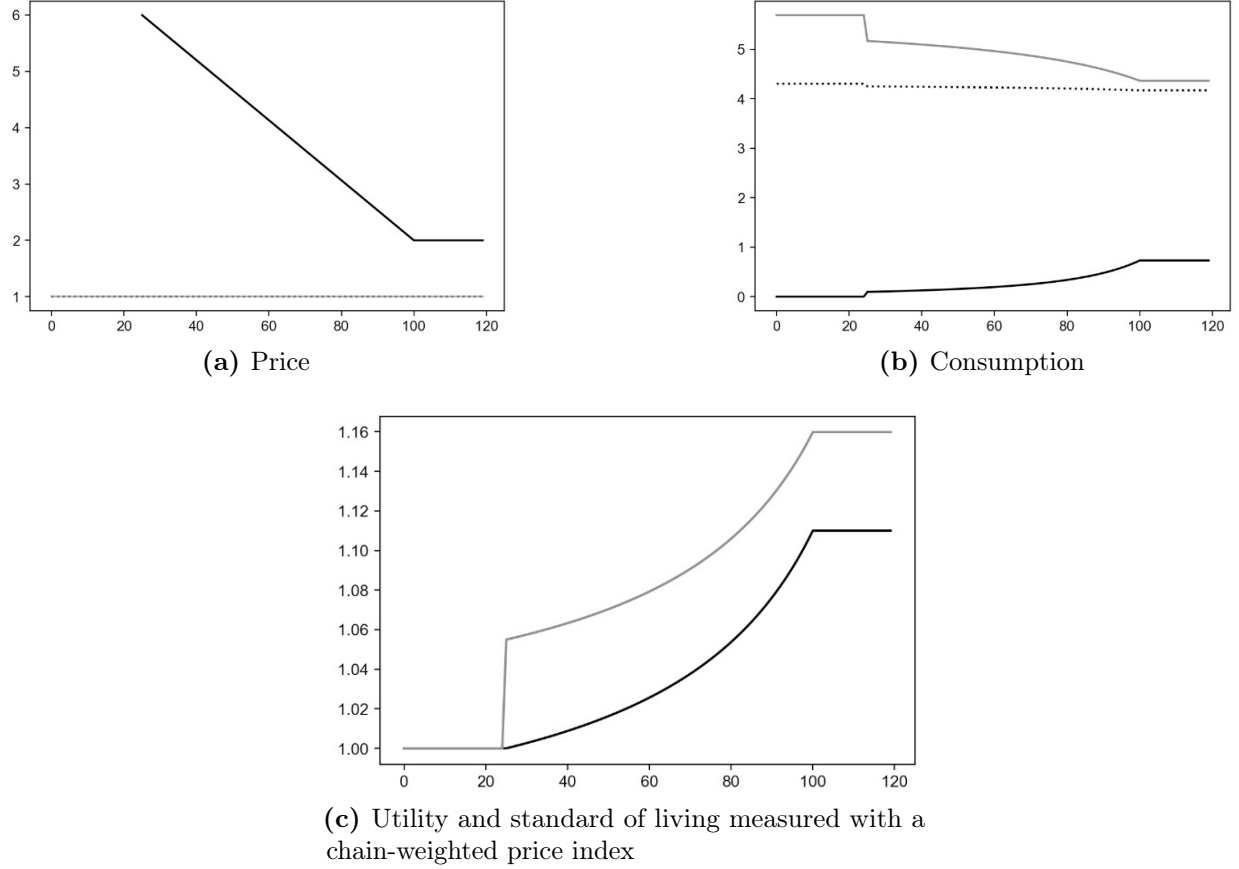
This result no longer holds if such homotheticity is not satisfied, as will be the case for specification (1). For example, if satisfaction reaches saturation more quickly for a particular good, the gain in well-being is not the same when all consumption is doubled from a state in which the good in question is already extensively consumed as when it is doubled from a state in which it is rarely consumed. This will be the case if B and/or B_0 are not equal to zero. In such cases, a discrepancy may arise between preferences and the standard-of-living measure based on a chain-weighted price index. One manifestation of this discrepancy is path dependence - the fact that the comparison of living standards between two dates t and t' depends on the path taken between the two periods; see Blanchet and Fleurbaey (2022) for a more detailed description. However, it remains useful to check that such discrepancies are limited in scale. This is the approach taken in this article.

With this point in mind, the use of this very simple model may invite three further objections, though they are not equally important:

- The first is that it appears to concern only a representative agent, with all the well-known limitations of that concept (Kirman 1992). Indeed, since green or brown goods may often be indivisible (e.g. electric versus non-electric cars), our simulations are more likely to represent a process of increasing adoption expressed through average weights in the basket of an aggregated “household” agent. However, the model’s qualitative insights remain translatable to the micro level and apply directly to divisible goods: the number of car trips rather than car ownership, lowering home temperature rather than changing the heating system, or reductions in any other consumption category with high greenhouse gas content. Everything presented can therefore inform the evaluation of the redistributive effects of

climate policies, in the spirit of Douenne (2020) and analyzes in terms of inflation inequality (Jaravel 2021).

- The second objection is that the reasoning is conducted in partial equilibrium. This limitation would indeed be very problematic if we aimed to offer comprehensive predictions of the effects of different greening policies. For example, the greening of preferences studied at the end of the article would have general-equilibrium effects on supply and demand and therefore on income and prices. These effects would need to be simulated to forecast their overall impact on well-being comprehensively. However, the objective of this article is more limited. If the net effects on income, prices and consumption patterns are directly observed by the statistician, the only question raised is whether the usual standard-of-living indicators synthesize them in a way that properly reflects their impact on household utility. This question essentially boils down to whether the nominal income deflators are valid, as illustrated by projections with constant nominal incomes. Only one feedback mechanism is simulated, in the case of taxation: the effect on nominal income of recycling tax revenue.
 - The final limitation is that the model neglects intertemporal effects. Whether individual or representative, the consumers we model choose their consumption pattern at each date based on current prices or constraints, without considering their future trends and without being able to smooth their responses to those trends. This is clearly a simplified view of a reality made more complex by anticipation and other intertemporal reasoning, especially when greening involves durable goods, with electric cars again serving as the canonical example: the higher purchase cost may be partially offset by lower running costs, although the ability to benefit from such compensation depends on financial assets, current resources or borrowing capacity, and potentially the resale value of the good. Here, the price of the good should be interpreted as an indicator of average usage costs over its service life, taking all these elements into account. When the green good is more expensive, it indicates that its lower energy consumption or maintenance costs are not sufficient to offset the higher purchase cost or the debt cost required for its purchase. A more explicit modeling of these intertemporal effects is beyond this preliminary overview, but Appendix A shows how they can be incorporated into a definition of the purchasing power of current income.
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Panels (a)-(b): All-purpose good — Good 1 — Good 2
 Panel (c): — Real income — Utility

Figure 1: The new good effect: the standard case.

Reading note: Good 2 appears on the market at date 25 thanks to an innovation that brings its price significantly below its reservation price. It is therefore immediately adopted to a significant degree, leading to a fall in consumption of good 1 and also favoring consumption of good 0. These changes have an immediate effect on the consumer's well-being that is not reflected in real gross disposable income. The latter nevertheless reflects the increase in well-being generated by the subsequent fall in the price of good 2 until date 100. If goods 1 and 2 were perfectly substitutable ($\sigma \rightarrow \infty$), the model would simulate a full and immediate switch from good 1 to good 2 as soon as the price ratio of the two goods exceeds the ratio of the services they provide $a/(1-a)$. In this case, deflation by chain-weighted prices would accurately account for all impacts on the consumer's standard of living: no impact while the price of good 2 remains above the switching price, a neutral change during the switch, followed by an increase in well-being linked to the subsequent fall in its price, which our indicators capture well. The only possible source of bias is the late introduction of good 2 into the basket of consumer goods considered by statisticians. Source: Authors' calculations.

In such a scenario, there is an initial positive impact on utility, as defined by equation (1), that is not captured by nominal income deflated by the chained price index. This limitation is well known in the measurement of prices and volumes: it is impossible to assess the initial impact of a new good when it first appears in the consumer basket if that first appearance occurs at a non-marginal level. However, the subsequent process by which a lower price for the good leads to a further increase in its consumption is well captured; it is indeed a process

of growth in which the gain induced by rising consumption of good 2 outweighs the declining consumption of good 1.

The underestimation of the impact of new goods highlighted by this initial simulation has been extensively discussed in debates about measuring the effects of information and communication technologies (ICT) and the digital economy.² This issue arises either when goods provide innovative services and capture significant market shares as soon as they are introduced, or when their inclusion in the price index is delayed until their market share begins to rise significantly. A similar underestimation could apply to the greening process, particularly for green technologies that are becoming increasingly competitive with polluting technologies. However, there is likely to be a significant difference from this optimistic version of creative destruction. Green goods typically do not offer new services but instead serve as alternatives to existing polluting goods, often at a higher initial cost. This initial extra cost may justify public intervention aimed at reducing the consumption of polluting goods. How these costs appear in standard-of-living measures depends on the method used to achieve greening.

Setting aside, for now, the case of voluntary sufficiency resulting from a change in preferences, we consider the two most frequently discussed options for “forced” greening. The first scenario involves introducing a tax on good 1, now assumed to be the polluting good (hereafter the brown good), with or without redistribution of the collected funds. The second consists in imposing a quantitative constraint on its consumption, an option often preferred when taxation meets too much resistance.

2.2 Forced Replacement Through Taxation or Regulation

If the incentive for greening takes the form of a Pigouvian tax, we remain within a framework governed by price signals. Unlike in the previous simulation, in which the price of good 2 fell, the price signal now consists of an increase in the price of the brown good. The expected impact is therefore a reduction in living standards, even if some tax revenue is redistributed, because of deadweight loss. The fact that redistribution cannot eliminate welfare losses is particularly easy to understand in the extreme case where the tax entirely eliminates consumption of the brown good: there would be no additional tax revenue to redistribute, while utility and the standard of living would obviously decline.

The impact of such a tax is simulated with and without revenue redistribution (Figure 2).³ Of course, this assessment does not take into account the utility gains expected to arise both in the long term and at a level broader than the local consumer. The purpose of the tax is to improve the long-term state of the world by taking into account externalities that market prices do not reflect. Here, however, the focus is on measuring the effect of the tax on the utility of a consumer who does not directly benefit from this improvement or is unaware of it. This effect is fairly well captured by the standard-of-living indicator. The slight discrepancy

²See, for example, Aghion et al. (2018) and, for literature reviews, Ahmad and Schreyer (2016) or Blanchet, Khder, et al. (2018).

³Revenue recycling is treated as lump-sum from the point of view of the household. The household takes the transfer as given when choosing its consumption bundle, even though, in equilibrium, the aggregate transfer is equal to aggregate tax revenue. This preserves the substitution effect of the tax while returning the revenue to households.

observed can be attributed to the non-homogeneity of the utility function, which causes a slight drift in the chained price index, but this bias is not material here.

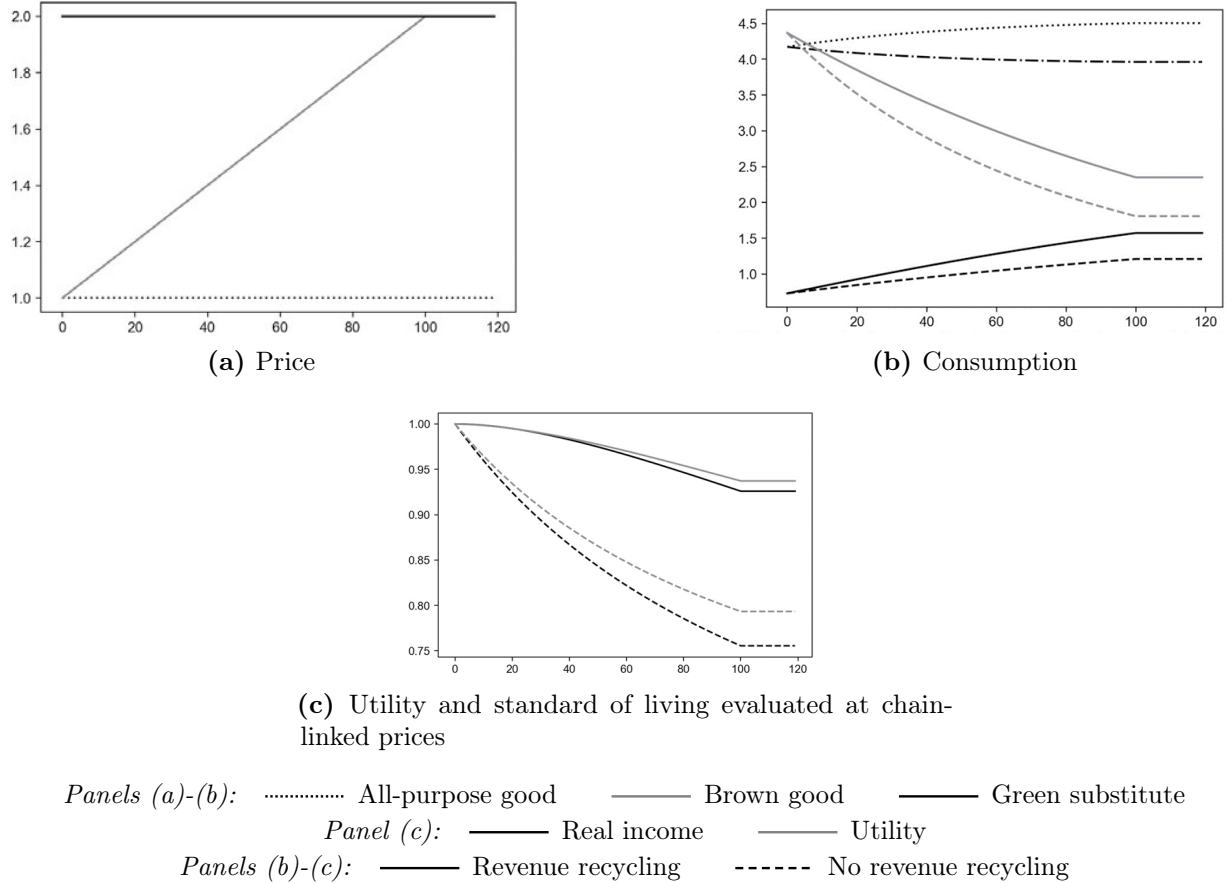


Figure 2: Impact of the taxation of the brown good on standard of living and well-being.

Reading note: The price of the green good (good 2) does not benefit from advances in greening technology. Greening results from an increasing tax on the brown good (good 1), with or without redistribution of the tax revenue. Redistribution does not prevent a decline in utility, though the decline is smaller than without redistribution. Income deflated by the chain-weighted price index takes account of this decrease in utility and of the fact that it is more marked in the absence of redistribution, even if the equivalence is only approximate given the non-homothetic nature of the utility function. Source: Authors' calculations.

What, then, would happen if, rather than taxing the brown good, the same consumption trajectory were achieved by a regulatory measure that reduces consumption of the brown good by the same amount but without any price signal? The overall consumption trend is identical to that under the taxation scenario with revenue recycling, as nominal income remains unchanged and allows the same opportunities to shift consumption towards the green good and the all-purpose good. Utility therefore evolves in the same way, again downward, but this time without any measured decrease in real income (Figure 3). Although the weights of the goods in the price index change, the index is unaffected in the absence of price changes, even though the increasingly restrictive regulation leads to the same continuous decline in utility that would occur with a price increase.

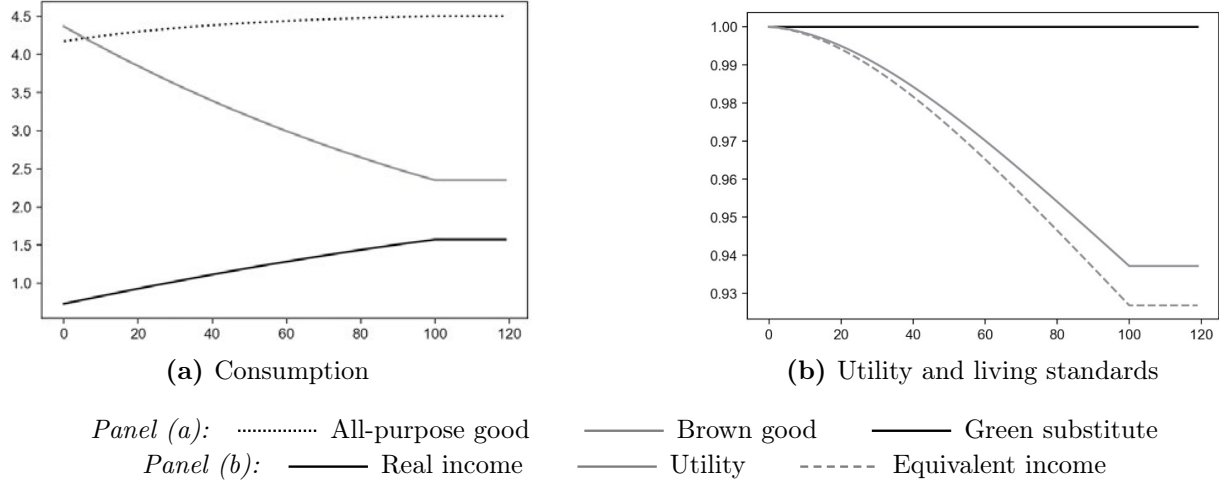


Figure 3: Impact of a regulatory constraint on the consumption of the brown good.

Reading note: The same reduction in consumption of the brown good is achieved by limiting its consumption through a regulatory measure. This results in a shift in consumption towards the green good, but also towards the all-purpose good. This forced change in the consumption structure leads to a fall in utility equal to that resulting from a Pigouvian tax with full revenue redistribution to households. However, no effect on Real income is recorded, since neither nominal income nor prices change. By contrast, evaluating the impact of the consumption constraint using the equivalent income method (Box 2) does capture the decline in utility, although a discrepancy remains because preferences are non-homothetic. Source: Authors' calculations.

The reason why a quantitative constraint is invisible to a conventional price index can also be expressed in terms of shadow prices. Consider the household problem

$$\max_q U(q) \quad \text{s.t.} \quad p \cdot q \leq R, \quad q_1 \leq \bar{q}_1.$$

Let λ be the multiplier on the budget constraint and $\mu \geq 0$ the multiplier on the regulatory cap. When the cap binds, the first-order condition for the brown good is

$$U_1(q) = \lambda p_1 + \mu = \lambda \left(p_1 + \frac{\mu}{\lambda} \right).$$

The term μ/λ is the implicit price increase that would decentralize the same reduction in brown consumption. Conventional price indices observe p_1 , but not the shadow price $p_1 + \mu/\lambda$. This is why a binding regulation can reduce welfare even when both nominal income and recorded market prices are unchanged.

Overall, while a significant share of the transition's effects on nominal income and its purchasing power is likely to be relatively well captured by national accounts - specifically those stemming from variations in nominal income or prices - some negative impacts may still be missed. These are the effects not automatically or fully translated into income and price signals. This issue touches on a classic critique of measuring living standards: the impact of non-discretionary consumption or mandatory expenditures, which limit the scope for consuming other goods. Purchasing requirements and bans are two facets of this problem. A more systematic approach to measuring living standards would incorporate them by calculating the income losses equivalent to such restrictions, assuming constant prices

(Box 2). Although the systematic application of these calculations in routine production may be difficult, we should at least be clear about what the measurement ignores at both macro and micro levels. Just as exposure to taxation or other sources of price variation can be unequal, the unequal impacts of regulatory measures should also be taken into account.

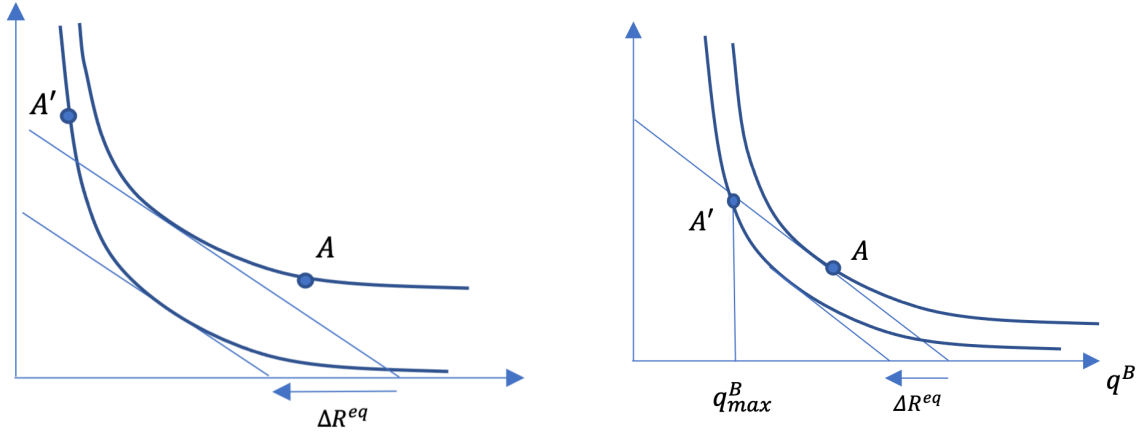
— **Box 2 - Monetary Valuation of the Impact of a Regulatory Constraint** —

From a theoretical point of view, valuing the monetary equivalent of a regulatory constraint can be seen as a specific application of the concept of equivalent income. Equivalent income provides a scalar ranking of consumption options that is consistent with ordinal preferences. It associates each utility level with the minimum income level required to achieve that level of utility under a given reference price system (Figure 4). Each reference price system corresponds to a set of parallel budget lines in the goods space. Utility isoquants are then cardinalized by the position of the line tangent to them. This position can be measured, for example, by the point at which this line intersects the horizontal axis, which amounts to treating the associated good as the numeraire. Formally, let the observed state be characterized by income R , prices p , a feasible set $\mathcal{C}$ and possibly non-monetary conditions z . Let p^{ref} , $\mathcal{C}^{\text{ref}}$ and z^{ref} denote the reference prices, feasible set and non-monetary conditions. The equivalent income associated with the observed state is

$$R^{\text{eq}}(R, p, \mathcal{C}, z; p^{\text{ref}}, \mathcal{C}^{\text{ref}}, z^{\text{ref}}) = \inf \left\{ m : \max_{\substack{q \in \mathcal{C}^{\text{ref}} \\ p^{\text{ref}} \cdot q \leq m}} U(q, z^{\text{ref}}) \geq \max_{\substack{q \in \mathcal{C} \\ p \cdot q \leq R}} U(q, z) \right\}.$$

In the case of a regulatory cap on the brown good, $\mathcal{C} = \{q \geq 0 : q_1 \leq \bar{q}_1\}$, whereas the unconstrained reference set is $\mathcal{C}^{\text{ref}} = \mathbb{R}_+^n$. The difference between R and R^{eq} can then be interpreted as the income loss that is equivalent, at the chosen reference prices, to the regulatory restriction.

This approach makes it possible to compare the utilities associated with two arbitrary points A and A' . The comparison is consistent with ordinal preferences, which standard-of-living indicators using chained prices - and still more those using fixed base-year prices - do not always guarantee. For example, in this figure, point A' would be considered better than point A if one used base prices corresponding to point A , since it lies above the corresponding budget line (dashed line), although it is on a lower isoquant. Chain-weighting reduces this risk of erroneous classification, but not completely, as the comparison of living standards between points A and A' may depend on the path taken to move from one to the other (path dependence).



Comparison of any two states A and A'

Capping of the consumption of the brown good

Figure 4: Equivalent income: general principle and application to a regulatory constraint.

This equivalent income method applies directly to cases where the move from A to A' does not result from a change in income or relative prices but from a cap on consumption of the good on the x -axis. The impact can be measured using the price system that the individual actually faces as the reference price system. The constraint forces the individual to reduce their consumption of the brown good below what they would ordinarily choose on the basis of income and prices alone. This leads to a shift towards the other good, assuming the individual exhausts their budget, but that shift would not be sufficient to keep utility unchanged, especially if the two goods are not easily substitutable. The variation in equivalent income accounts for this.

This approach resembles another equivalence calculation for climate policy instruments: the economy-wide carbon price equivalent (ECPE) used by the IMF (Black et al. 2022). ECPEs correspond to the level of carbon tax that would result in the same reduction in emissions as the policy or policies under consideration, which may include standards or constraints. Although the two approaches may overlap, they should not be conflated. In the carbon-price-equivalent approach, two policies are considered equivalent if they lead to the same reduction in emissions. In this article, we use equivalent income to determine whether these policies have the same effect on well-being. Two policies can be equivalent in both senses if they both lead to the same mix of brown, green and generic consumption: in that case, the same reduction in emissions corresponds exactly to the same variation in utility. However, this overlap is not guaranteed. For example, a non-recycled tax and a constraint that both lead to the same reduction in brown consumption would be equivalent according to the ECPE, but not in terms of their effect on well-being, since non-recycling of the tax revenue would generate a negative income effect for the consumption of all goods. More generally, the divergence may therefore provide a criterion for choosing between decarbonization options: for a given environmental impact, priority should be given to the policy that is least detrimental to well-being. Moreover, from a social well-being perspective that takes inequalities into account, an additional difference is that the various options do not affect everyone in the same way: the same overall reduction in emissions can be achieved with varying degrees of equality.

3 Non-Monetary Co-Benefits

The effects discussed in this first overview appear rather negative. While income and price statistics should spontaneously capture anything transmitted through these two variables, they risk missing the effects of regulatory constraints. This limitation deserves special attention: underestimating the costs to households could result in poor anticipation of resistance to change and inadequate sizing of the measures needed to make the transition more bearable.

However, could other factors work in the opposite direction? There are two aspects to this question. The first involves identifying the possible non-monetary effects of the transition, some of which could be favorable. These effects must be listed and assessed to determine the extent to which they might offset the monetary costs. The second aspect concerns the possibility that the transition may be accompanied by changes in preferences: everything discussed so far has been based on the implicit assumption that preferences for polluting and green goods are stable. Yet an additional vector of greening is voluntary sufficiency (Pommeret et al. 2023; Oliu-Barton et al. 2024), that is, a change in preferences in favor of green goods or even towards reduced overall consumption.

These two aspects partially overlap: preferences may shift in a way that reduces the importance of polluting goods while increasing the weight of green goods, and may also give more weight to the non-monetary co-benefits of the transition. For clarity, we separate these two topics, starting with the evaluation of the non-monetary impacts of the transition if preferences were to remain stable. In doing so, we do not claim to provide a systematic inventory of these co-benefits but limit ourselves to a few methodological observations.

First, with regard to the list of these non-monetary effects, it should be recalled that our focus is on current living standards and well-being. We therefore consider only co-benefits that have sufficiently immediate effects to offset equally immediate costs. Longer-term benefits belong to the question of sustainability. In addition, the non-monetary effects of the transition are not necessarily all co-benefits; some may actually exacerbate costs. Positive impacts include immediate gains in health, leisure and the quality of the living environment. Many negative effects can nevertheless also be anticipated. For example, restructuring caused by the transition to greener production will result in a mix of job losses and job creation, potentially leading to periods of unemployment and/or transitions to different types of work (Hentzgen et al. 2023). Changing jobs can affect well-being beyond its effects on income; this is even more true for those who experience unemployment, since the loss of well-being associated with unemployment is greater than the difference between previous wages and unemployment benefits. Another example is the potential need for increased housing density as part of a shift towards greater sufficiency. Income per consumption unit could record this positively because of larger economies of scale from shared living arrangements, but this would not correspond to actual well-being: the historical trend towards reduced cohabitation suggests that individuals are willing to sacrifice purchasing power for the benefits of living alone, benefits they may be forced to give up.

From a methodological point of view, whether these non-monetary effects are positive or negative, the question remains the same: how can they be integrated with monetary indicators? This is the question traditionally addressed in the “beyond GDP” literature, which proposes four options for incorporating these non-monetary factors into the measurement of well-being, as previously outlined in Blanchet and Fleurbaey (2020).

The first approach is the use of dashboards, which entails presenting a range of indicators that shed light on various aspects of living conditions and well-being without attempting to aggregate them. For instance, in their comparative analysis of various avenues for achieving greener consumption with potential positive effects on well-being, Creutzig et al. (2021) organize these effects according to the categories of the Sustainable Development Goals currently promoted by the United Nations. However, the problem with this approach is the sheer volume of information it generates and the difficulty of prioritizing it. Synthesized information is also needed. In the end, when dashboards are used to make trade-offs between policies with different effects on different aspects of well-being, decisions ultimately rely on some form of implicit aggregation and on non-transparent hierarchies among these dimensions.

The second approach is that of composite indices. It merits mention given its long-standing prominence in the search for alternatives to GDP. It uses a number of techniques to make otherwise incommensurable items statistically comparable, and then aggregates them into a single index according to conventional rules. The method is considered transparent because the aggregation rules are based on fairly basic arithmetic. Its limitation lies in the fact that the resulting relative valuations may not reflect individual preferences or relevant social choices, since they are the mechanical result of a purely statistical aggregation rule.

By contrast, one advantage of the third approach - the measurement of subjective well-being - is its apparent respect for individual preferences. It eliminates the need to formulate principles for aggregating different components of well-being, relying instead on what individuals report about their overall well-being, using a cardinal approach. Individuals indicate how favorable they perceive their living conditions to be, typically by scoring their perceived overall well-being on a scale of 0 to 10, without needing to make their personal weighting of different well-being dimensions explicit. The appeal of this method lies in the fact that it goes directly to the outcome of interest, while also allowing the unequal distribution of subjective well-being to be taken into account. This is something that composite indicators based on macro data are unable to do. Even when they try to include inequalities measured across different dimensions, composite indicators fail to capture the cumulative impact of deprivations when they are correlated across dimensions. These advantages make the method particularly useful for addressing many questions, and it is a natural candidate for assessing the “all-encompassing” impact of climate transition (Perona 2022).

However, relying on self-reported well-being raises the problem of the relativity of the scales on which individuals assess their situation. The fact that an individual A could feel less happy than another despite having the same circumstances is certainly interesting to measure. Problems arise, however, when this leads to inconsistencies with ordinal preferences, which is precisely what we are trying to avoid. For example, an individual A might prefer their current situation to that of another individual A' , yet rate their own well-being less favorably than A' if they are naturally more demanding than A' . It would be manifestly wrong to conclude that a society composed predominantly of individuals of type A is worse off than if it were composed primarily of individuals of type A' . The method risks producing results that conflict with individuals’ actual ordinal preferences.

A fourth approach addresses this problem, and has already been introduced in Box 2 in the context of calculating a monetary equivalent for rationing of the polluting good. The method of equivalent income allows a monetary translation of anything that affects the ability to generate utility from income, with prices and rationing constraints being only specific

cases. The situation where utility is affected by both prices and one or more non-monetary factors is discussed in Box 3. The approach involves setting reference values for prices as well as for the non-monetary factor(s), and then calculating the income that would allow the current level of utility to be achieved in the hypothetical situation in which the individual is exposed to these reference values rather than to their current values. For instance, if an individual with income R is in a poor state of health H and if good health H^{ref} is taken as the reference, their equivalent income R^{eq} will be the amount that guarantees the equality $U(R^{\text{eq}}, H^{\text{ref}}) = U(R, H)$. This income would be lower the worse the health state H is, while it would increase if an improvement in the environment led to better health. The method provides a monetary valuation of this improvement that is consistent with ordinal preferences over health and monetary living standards. It is rooted in the tradition of cost-benefit analysis.

However, the problem posed by this method lies in its implementation. Several techniques, or combinations of techniques, are possible: relying on preferences revealed through behavior; using contingent valuation techniques, i.e. directly asking individuals how much they would be willing to pay or accept for specific changes in their situation or environment; or combining objective data with the subjective satisfaction measures discussed above. The idea is to measure how individuals are willing to trade material factors off against other aspects of their living conditions by empirically analyzing how these factors affect subjective well-being. This can be done through surveys that combine direct measures of perceived well-being with objective components. Utility-function calibrations from the existing literature may also be used.

The cost of implementing these different techniques makes it difficult to imagine applying them in routine statistical production. Among the practical applications of this approach, one notable example is the work of Serres and Murin (2016), who attempt to calculate the extent to which increased life expectancy associated with a reduction in local pollution - resulting from decreasing greenhouse gas emissions - could offset the economic costs of these reductions. They find that the compensation is only partial, but this does not exhaust the topic, as their focus is limited to only one co-benefit of the transition.

— **Box 3 - Equivalent Income Applied to Non-Monetary Co-Benefits** —

To preserve the possibility of a two-dimensional representation, we assume that preferences depend only on the capacity of income R to purchase a composite all-purpose good q and on a single non-monetary co-benefit described by the variable z : this could be, for example, the state of the environment or health as affected by the state of the environment. If the composite good is used as the numeraire, the method requires only a reference value z^{ref} for the variable z . The equivalent incomes associated with states A and A' correspond to the x -coordinates of the points at which the isoquants and the horizontal line $z = z^{\text{ref}}$ intersect. In this example, income decreases between state A and state A' , but this is more than offset by the increase in z .

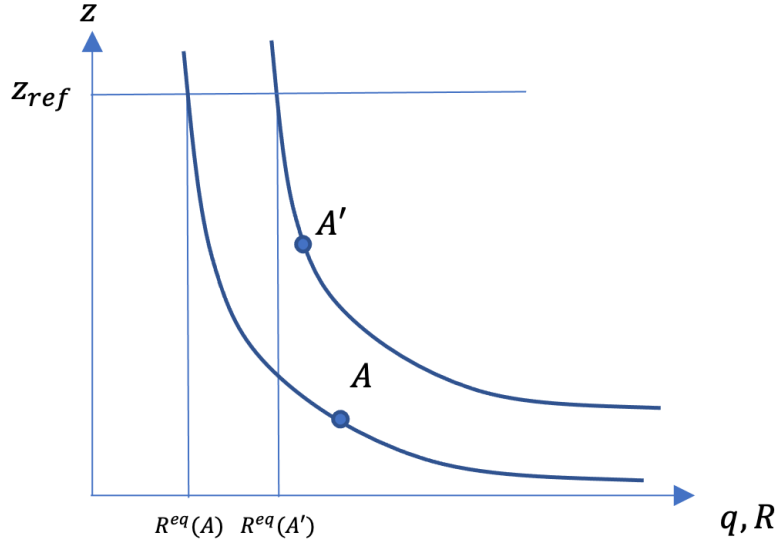


Figure 5: Equivalent income when well-being depends on a market good and a non-monetary factor.

It should be noted that, in practice, some well-being effects associated with improvements in health may already be measured by some of our usual indicators through the associated reduction in healthcare expenditure. Gross disposable income should indeed capture reductions in health-insurance spending if they lead to a reduction in the taxes that finance it. However, the effect of lower medical expenses paid by households, which would free up income for other expenses, would be taken into account only through a discretionary-income approach that treats these medical expenses as constrained spending. Finally, the so-called “adjusted” disposable income approach, which integrates the monetary equivalent of individualized transfers, would in principle miss all these effects, since the decrease in taxes would be fully offset by the decrease in health benefits.

4 Transition and Well-Being with Changing Preferences

Even without co-benefits such as those described in the previous section, the cost of the transition in terms of perceived living standards or well-being could still be reduced if the transition is accompanied or driven by a greening of preferences. If meat consumption or air travel falls sharply simply because individuals spontaneously stop eating meat or traveling by plane, their well-being will not be negatively affected, regardless of what happens to prices or regulations.

But how can one quantify the dampening effect of such changes in preferences? Attempting to measure the cost of living or real income growth in the context of changing preferences is like trying to compare the size of different objects with an elastic ruler that expands or contracts as one moves from one object to the next. There seems to be no solution to this

problem, which is undoubtedly why, until recently, it was largely ignored by the literature (Samuelson and Swamy 1974; Balk 1989), despite the fact that past growth has clearly been accompanied by radical changes in preferences.

The COVID-19 crisis gave researchers a first reason to stop ignoring this subject: it drastically changed preferences over different types of goods and services, with some goods suddenly becoming essential and others dispensable (Baqae and Farhi 2020; Baqae and Burstein 2021; Blanchet and Fleurbaey 2022). This raised two questions: how can standard statistical indicators that implicitly rely on stable preferences be interpreted in such a context, and which actual measures of utility or well-being can we try to compare them with? These two questions are even more critical in the context of the green transition, since changes in preferences are no longer an exogenous disruptive factor that we can afford to ignore for lack of a better option. In the green transition, we expect them to play an active role (Konc, Savin, and Bergh 2021; Mattauch, Hepburn, et al. 2022; Mattauch, Stern, and Konc 2022), and policies, including sufficiency policies, are being considered to encourage them. Continuing to ignore them is increasingly untenable. The two approaches just presented for handling non-monetary co-benefits — subjective well-being measurement and the equivalent income method — each offer a way to address this issue, and both deserve consideration.

One could choose to rely on the subjective measurement of well-being. For example, in the presence of taxation, it would be reasonable to assume that the subjective indicator would incorporate both the negative impact of the tax and the fact that individuals quickly learn to put its effects into perspective, or infer information about the consequences of climate change and integrate it into their preferences. More generally, one could even say that the very nature of these subjective indicators allows them to take account of all possible ways in which preferences may change. This is how we are accustomed to interpreting the best known of their stylized messages, the Easterlin paradox (Easterlin 1974), which states that beyond a certain stage of development, economic growth has only a minor impact on subjective well-being. One possible explanation of this paradox is that needs and aspirations increase as material living conditions improve, and that the gap between living conditions and those aspirations determines declared well-being. This is not necessarily a change in ordinal preferences, but at the very least a change in the way they are translated into cardinal terms.

If such an explanation of the Easterlin effect is correct, it could play out in the opposite direction along a green transition path: chosen sufficiency would involve a slowdown or even a decline in consumption at the same time as growth slows or declines, thereby limiting the fall in subjective well-being. This calls for careful examination of how these subjective indicators would behave along a transition trajectory with changing preferences.

We can also examine what equivalent-income measures of the standard of living introduced earlier would say. What, first, can be said about the impact of these preference changes on the usual standard-of-living indicator, calculated as nominal income deflated by a chained price index, given that statisticians will continue to calculate it anyway? If preferences remain constant, the tax generates a substitution effect that mitigates some of the impact that would have occurred if substitution were not an option. In principle, that impact is taken into account by the chained index, although the path dependence introduced by chain-weighting can still be an issue, even with constant preferences. The greening of preferences would strengthen this substitution effect, adding a second factor that moderates the effects of the tax, while also introducing a second path-dependence issue. Consider an initial state A with

brown preferences, a final state A' with greener preferences, a tax increase between these two periods, and two different scenarios for the trajectory of preferences: one in which greening occurs before the tax increase and another in which it follows the tax increase. It is clear that the message conveyed by chain-weighted indices would not be the same in these two scenarios: in the first case, the increase in the price of the brown good would be reflected in the price index with a weight that had already begun to decrease, whereas this would not be the case in the second scenario. The fall in the standard of living would therefore be judged smaller along the first trajectory than along the second, even though both trajectories have the same starting and ending points.

Thus, while the standard indicator need not be silent on a potential moderating effect of changing preferences, its message will be partial and unstable. Would the equivalent income method help avoid this problem? From a technical point of view, nothing prevents its implementation, as it does not require considering individuals with identical preferences (Fleurbaey and Tadenuma 2014). There is no need to assess how each individual would feel with preferences that are not their own. Instead, each individual uses their own preferences to assess the hypothetical level of income that would make them indifferent between their current state and a situation in which they faced the chosen reference price system. Comparisons are then made based on these equivalent incomes, whether they are interpersonal comparisons between two individuals living at the same time or comparisons over time for the same individual with two successive preference systems.

Uncertainty about the final message is not entirely resolved, however; it takes a different form: the result of the comparison depends on the chosen reference price system. That dependence is shown in Box 4 and can be illustrated by a final simulation using the article's three-good model (Figure 8). This simulation also allows for a comparison with the evolution of standard-of-living indicators measured with a chain-weighted price index, in the context of one of the two scenarios in Figure 2: the introduction of a tax without recycling its revenue - which arguably represents the least favorable situation for consumers - coupled with preferences that become greener as the tax increases. The change in preferences involves a joint change of parameters B and a of the utility function from -1 to -2 and from 0.55 to 0.25 , respectively, between dates 0 and 100. The fall in B indicates a reduced need to consume the service provided by both the green and the brown good (for example, less need to travel by car, regardless of whether that car is electric), and the fall in a means that, in producing this service, an increasing preference is assigned to good 2, the green good.

— Box 4 - Equivalent Income with Variable Preferences —

We consider a green good and a brown good together with a change in preferences in favor of the former, assuming for now that income and prices remain unchanged (Figure 6). Initial and final preferences are represented by the functions U and U' . The change in preferences leads to a shift from point A to point A' . The classification of these two points using the equivalent income method depends on the reference price system: point A appears above point A' with the reference prices in the left-hand figure and below it with the reference prices in the right-hand figure. This indeterminacy may appear irresolvable, but it simply reflects the fundamental indeterminacy created by the change in preferences: A is preferred to A'

by an individual with the initial preferences, and A' is preferred to A by an individual with the final preferences. In cases where prices remain the same in both states, these prices can be used as reference prices, in which case the two states will be judged equivalent. This is ethically relevant: two individuals with different preferences but the same income at given prices are considered equally well off, even if their consumption choices differ.

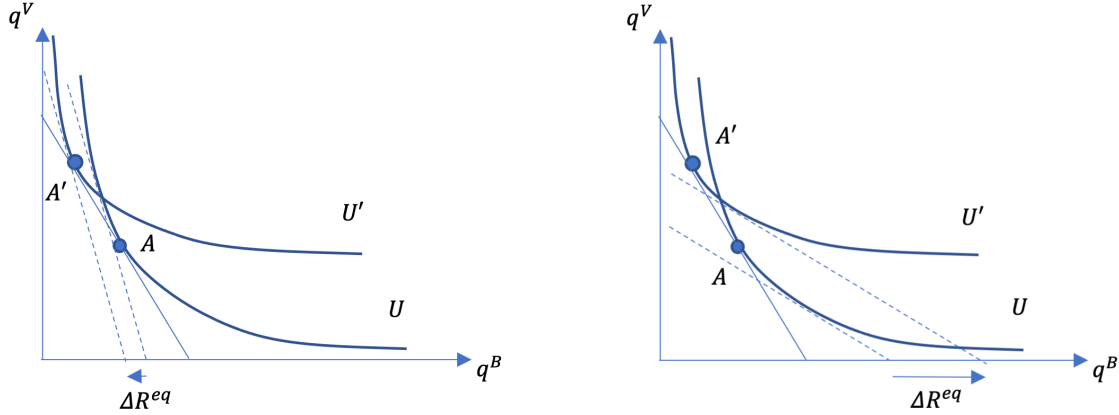


Figure 6: Effect of a change in preferences, at given prices and income, depending on the chosen reference price system.

This solution is no longer available when the change in preferences is accompanied by a change in prices. This would be the case in particular if the greening of preferences goes hand in hand with the introduction of a tax on the brown good. Consider a two-stage process (Figure 7) in which the tax first pivots the budget constraint and leads to a shift from point A to point A' with preferences unchanged, after which a change in preferences results in a shift to point A'' without any change to the budget line.

If final prices are used as the reference prices (right-hand figure), the transition from point A' to point A'' has no effect on equivalent income, and only the decrease in equivalent income between point A and point A' is observed. In other words, the compensatory effect of the preference change is not accounted for. By contrast, some compensation is measured if initial prices are chosen as reference prices (left-hand figure), resulting in a back-and-forth movement that remains partial in the figure but would be complete if the change in preferences were such that, ultimately, the individual entirely forgoes the brown good.

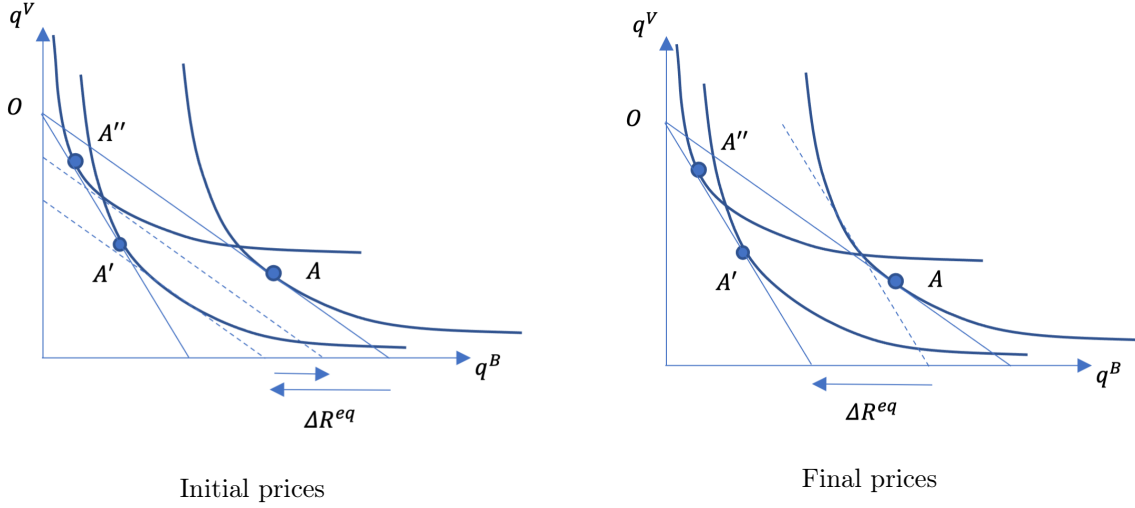


Figure 7: Effect of environmental taxation on equivalent income with variable preferences, with reference prices equal to the initial prices or final prices.

We have focused here on a pure change in preferences that necessarily goes in the direction of greener behavior. This calls for three remarks:

- This type of greening should be distinguished from greening that results from a pure income effect with no change in preferences, as would be the case if environmental quality were a superior good to which consumers attach greater importance as their income rises. In this case, we would remain within a fixed-preference framework. However, this scenario still encounters challenges due to the non-homothetic nature of preferences, which complicates the interpretation of real-volume or standard-of-living indicators in terms of utility.
- the standard for well-being is households' true preferences, which must be distinguished from so-called behavioral preferences, i.e. those revealed by consumers' actual choices (Farhi and Gabaix 2020). A discrepancy arises between the two when individuals fail to internalize all the personal consequences of their choices, typically because of a lack of information. If true preferences are stable and only behavioral preferences change, for example as a result of a nudge that corrects the discrepancy between the two, we remain within a fixed-preference framework, and the effect of the nudge is generally positive. For further reading, see Pommeret et al. (2023).
- Changes in true preferences can occur spontaneously, be influenced by communication campaigns or be driven by peer effects. They may also result from the introduction of taxes or regulatory changes that increase awareness of environmental issues, thereby enhancing their effectiveness (Konc, Savin, and Bergh 2021). However, opposite reactions may also be observed, where taxation or constraints lead to backlash (Ehret et al. 2022).

We compare the evolution of income deflated by a chain-weighted price index with two versions of equivalent income, one using initial prices as the reference and the other using current prices at each date (Figure 8). Unsurprisingly, the decrease in deflated income is smaller than in the case of taxation with fixed preferences shown in Figure 2. This is

because the weight of the taxed good in the index decreases faster than it would with fixed preferences, resulting in a smaller deterioration in living standards, although this offset will differ depending on whether preferences change before or after the tax is introduced.

The equivalent income approach avoids this form of path dependence since, at each date, it involves only current preferences, regardless of how they have evolved since the start of the process. However, the outcome depends on the prices used as the reference. When the reference prices are those of the initial period (upper trajectory), equivalent income evolves under the influence of two contradictory forces: the tax increase logically reduces equivalent income, but the change in preferences has a positive impact since it allows the consumer to move away from a good that has become expensive compared with its price in the reference system, namely the initial untaxed price. Here, this second effect prevails over the first, since equivalent income changes its slope as soon as preferences start to become greener. At the end of the transition, equivalent income remains below its initial value, but in the limiting case where the consumer becomes fully “green” and no longer wishes to consume the brown good at all, even at its initial price, their equivalent income would return to its initial value because they would have become completely indifferent to the price of the now-undesired brown good. Under the other convention, using current prices as the reference system (lower trajectory), the effect of the change in preferences is fully neutralized in current equivalent income, since it is equal to current income by definition. However, in each period, the change in reference prices updates the value of initial equivalent income, which increases because the initial situation appears retrospectively more advantageous as the reference price of the brown good increases: compared with the after-tax price, brown consumption was initially implicitly subsidized. As a result, when compared with an increasing initial equivalent income, current equivalent income appears to be in continuous decline.

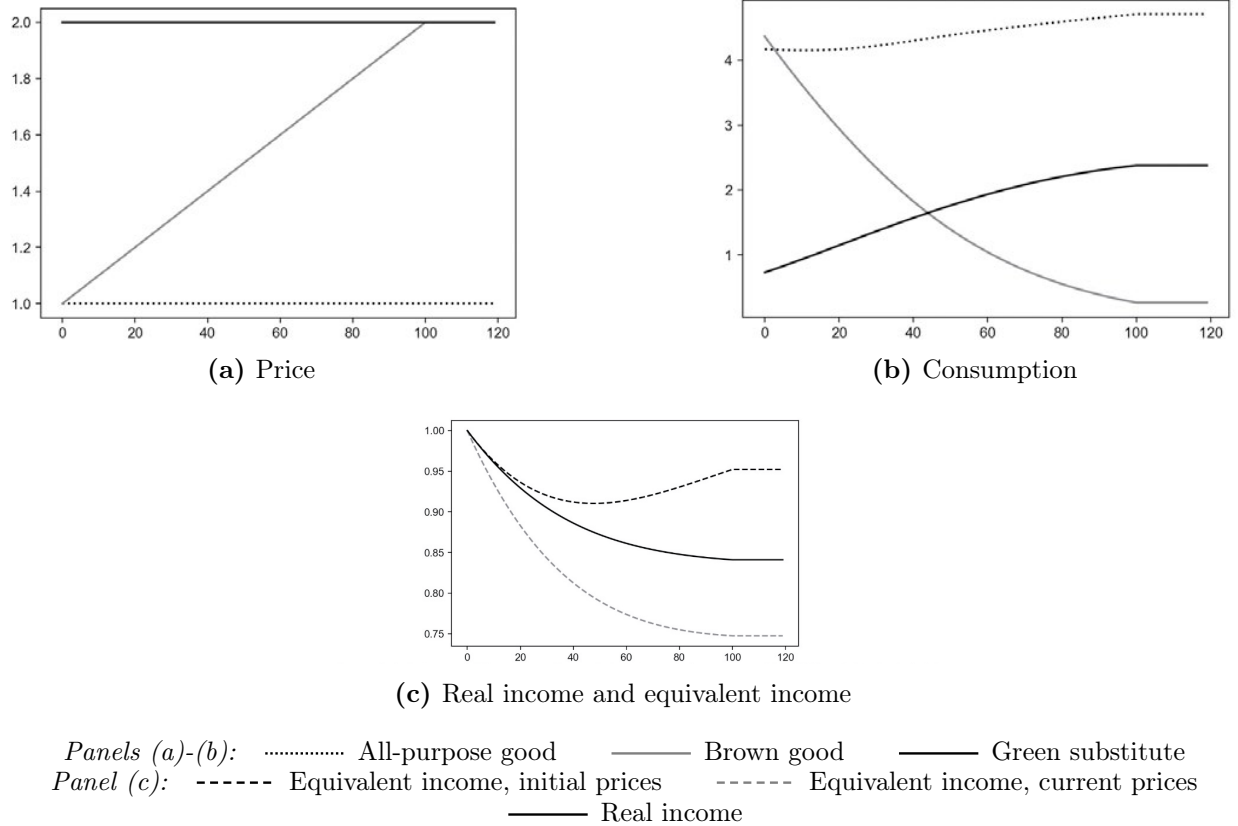


Figure 8: Scenario involving taxation of the brown good (without recycling of the tax revenue) coupled with a greening of preferences.

Reading note: Based on the same initial values as in the stable case, the parameters B and a shift linearly from -1 to -2 and from 0.55 to 0.25 between 0 and 100 . The decline in consumption of the brown good is more pronounced than in the simulation depicted in Figure 2 with an identical tax trajectory and without recycling. The impact on equivalent income is evaluated using either initial prices or current prices as the reference prices. Source: Authors' calculations.

In the end, there are two different and complementary points of view on the changes at play, and they bracket the evolution of real income deflated by a chained price index. One perspective suggests that changes in preferences compensate for costs, although this viewpoint cannot be entirely privileged either. The measurement of living standards was already affected by unavoidable perspective effects with non-homothetic but stable preferences; this problem can only be amplified when preferences are unstable.

5 Conclusion

The answer to the question asked in this article is mixed. Standard income and price statistics are not obsolete in the context of the climate transition. When transition effects operate through observed prices, disposable income or transfers, existing real-income indicators remain informative. This is the case for many forms of green creative destruction and for carbon taxation, at least in their immediate effect on households. The main difficulties arise at the boundary of what prices and incomes record: binding quantitative regulations can reduce

welfare without changing either nominal income or observed prices; non-monetary co-benefits may offset part of the transition cost without being captured by ordinary income deflators; and endogenous greening of preferences may make the welfare cost of lower brown consumption smaller, while making real-income comparisons more path-dependent and normatively more fragile. Table 1 summarizes these distinctions.

Table 1: Transition mechanisms and their treatment by standard real-income indicators

Mechanism	Main welfare channel	Captured by standard real income?	Main residual issue
Green creative destruction	Falling relative price of green goods	Largely yes, once the good is included in the price index	New-good bias, non-homotheticity
Carbon taxation	Higher observed price of brown goods, possibly with transfers	Largely yes	Distributional incidence, non-homotheticity
Quantitative regulation	Direct restriction of the feasible consumption set	No, unless the shadow value of the constraint is reflected in observed market prices	Need for shadow prices
Non-monetary co-benefits	Health, local environmental quality, leisure, employment quality...	No, except indirectly when they affect income, taxes or expenditures	Monetary valuation of non-market dimensions
Changing preferences	Lower desired consumption of brown goods or weaker material consumption needs	Partly, through changing expenditure weights in chain indices	Path dependence and normative ambiguity

The appropriate response is therefore not to replace standard indicators, but to complement them. A statistical system suited to the climate transition should combine conventional measures of real household income with explicit information on shadow costs, non-monetary co-benefits, changing preferences and their distribution across households. This ‘current well-being’ perspective should itself remain complementary to sustainability indicators, whose purpose is to assess whether present policies are sufficient to protect future living conditions.

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A Appendix: Purchasing Power under Intertemporal Choices

This appendix extends the discussion in Box 1, where intertemporal choices were deliberately set aside. The extension is relevant because several central instruments of the climate transition involve goods purchased at one point in time, but whose use is spread over time (electric vehicles, heating systems, home insulation, solar panels, energy-efficient appliances...). This raises a natural question for the construction of price indices: should the purchase price be recorded at the time of acquisition, or amortized over the full service life of the good? This appendix addresses that question.⁴

To do so, it first addresses a more general issue: how should the intertemporal nature of consumption choices be taken into account when defining real income? Two concepts are commonly used in the literature:

1. *Permanent income*, which is a wealth-like concept. It includes current income, expected future income, and accumulated assets, and is therefore suited to measuring intertemporal well-being as a whole.
2. *Real current income*, typically defined as the maximum quantity of current goods and services that nominal income can purchase; equivalently, it tracks the evolution of the static indirect utility

$$\max_x U(x) \quad \text{subject to} \quad px \leq I. \quad (2)$$

Neither concept, however, captures exactly the object needed here. We call this object the *purchasing power of current income*. It asks what the income received at date t contributes to intertemporal well-being when it may either be spent immediately or saved for future consumption. It lies between the two concepts above. It is narrower than permanent income because it isolates current income rather than total lifetime resources. But it is broader than static real current income because current income may finance future consumption and the services of durable goods. The purpose of the appendix is to make this object explicit and to show what its deflator should contain.

Notation

We use the following notation throughout:

x_t Vector of non-durable consumption at date t .

$U(x_t)$ Instantaneous utility at date t .

$W((U_s)_{s \geq t})$ Recursive intertemporal welfare over the sequence of period utilities.

I_t Nominal income at date t .

⁴The contribution of durable goods to living standards also depends on the operating and maintenance costs incurred during their use. These costs are important empirically but they do not raise a specific measurement issue related to durability: they are ordinary period costs and can be included in the relevant period price vector.

p_t Vector of prices for non-durable consumption at date t .

r_t Nominal interest rate applying to savings between t and $t + 1$.

$\widehat{I}_{s|t}, \widehat{p}_{s|t}, \widehat{r}_{s|t}$ Expectations, formed at date t , of income, prices, and interest rates relevant for future dates $s > t$.

k_t Financial wealth carried from previous periods, expressed in nominal terms at date t .

$\widehat{R}_{s|t}$ Future value in period s of one unit of currency saved at time t :

$$\widehat{R}_{s|t} \equiv \begin{cases} 1, & s = t, \\ \prod_{m=t}^{s-1} (1 + \widehat{r}_{m|t}), & s > t, \end{cases} \quad \text{with } \widehat{r}_{t|t} \equiv r_t.$$

$\mathcal{P}_t$ Price index at date t (unit-expenditure index) defined below.

$\widehat{\mathcal{P}}_{s|t}$ Price index associated with expected prices at date s .

For each date $s \geq t$, define the date- t price of one unit of period- s utility by

$$\pi_{s|t} \equiv \frac{\widehat{\mathcal{P}}_{s|t}}{\widehat{R}_{s|t}}. \quad (3)$$

With the conventions

$$\widehat{\mathcal{P}}_{t|t} = \mathcal{P}_t \quad \text{and} \quad \widehat{R}_{t|t} = 1,$$

we have $\pi_{t|t} = \mathcal{P}_t$. For $s > t$, $\widehat{\mathcal{P}}_{s|t}$ is the expected nominal cost of one unit of period- s utility, while $\widehat{R}_{s|t}$ converts that future cost into date- t units. Let

$$\boldsymbol{\pi}_t \equiv (\pi_{s|t})_{s \geq t}.$$

A.1 Intertemporal environment

Consumers have homothetic per-period preferences over goods represented by a homogeneous utility function $U(\cdot)$, and intertemporal preferences over sequences of utility levels represented by a recursive functional W . Specifically, there exists a function f such that, for any date t ,

$$W((U_s)_{s \geq t}) = f\left(U_t, W((U_s)_{s \geq t+1})\right). \quad (4)$$

The recursive notation is only a convenient way of saying that the consumer evaluates today's utility and the continuation value of future utilities jointly. It also makes clear how saving enters the problem: saving is valuable because it raises the continuation value, not because it is a final good in itself.

For each period s , let $e(p_s, U_s)$ denote the expenditure function associated with U , i.e.

$$e(p_s, U_s) \equiv \min_{x_s} \{p_s x_s : U(x_s) \geq U_s\}. \quad (5)$$

Under homotheticity of U , the cost of a given utility level is proportional to that level: $e(p_s, U_s) = \mathcal{P}_s U_s$, where $\mathcal{P}_s = e(p_s, 1)$ is the cost of one unit of instantaneous utility at date s . For expected prices $\hat{p}_{s|t}$, we similarly define

$$\hat{\mathcal{P}}_{s|t} \equiv e(\hat{p}_{s|t}, 1).$$

Thus, after aggregating goods within each period, the consumer can be viewed as choosing a sequence of utility quantities (U_s) , each of which has a price $\mathcal{P}_t$ or $\hat{\mathcal{P}}_{s|t}$. This is the step that allows the intertemporal problem to be written as a standard choice over dated utility flows.

The consumer's problem at date t

At date t , the consumer observes current nominal income I_t , current prices p_t , the current interest rate r_t , and financial wealth k_t . The consumer also forms expectations about future income, future prices, and future interest rates. Given this information, the consumer chooses a consumption path $(x_s)_{s \geq t}$ to solve

$$\max_{(x_s)_{s \geq t}} W((U_s)_{s \geq t}) \quad (6)$$

$$\text{subject to } U_s = U(x_s) \quad \text{for all } s \geq t,$$

$$p_t x_t + \sum_{s > t} \frac{\hat{p}_{s|t} x_s}{\hat{R}_{s|t}} \leq I_t + \sum_{s > t} \frac{\hat{I}_{s|t}}{\hat{R}_{s|t}} + k_t. \quad (7)$$

The left-hand side is the present value, at date t , of the planned consumption path. Current consumption is paid for at current prices. Future consumption is valued at expected future prices and discounted by the expected return on saving. The right-hand side is total available resources at date t : current income, the present value of expected future income, and wealth brought into the period.

Using the expenditure representation (5), we can re-express the problem in terms of utility levels U_s instead of consumption bundles x_s . The minimal cost of achieving utility U_s at date s is $\hat{\mathcal{P}}_{s|t} U_s$, so the intertemporal budget constraint becomes

$$\mathcal{P}_t U_t + \sum_{s > t} \frac{\hat{\mathcal{P}}_{s|t}}{\hat{R}_{s|t}} U_s \leq I_t + \sum_{s > t} \frac{\hat{I}_{s|t}}{\hat{R}_{s|t}} + k_t. \quad (8)$$

The problem is therefore equivalent to

$$\max_{(U_s)_{s \geq t}} W((U_s)_{s \geq t}) \quad (9)$$

$$\text{subject to } \sum_{s \geq t} \pi_{s|t} U_s \leq I_t + \sum_{s > t} \frac{\hat{I}_{s|t}}{\hat{R}_{s|t}} + k_t.$$

Equation (9) is the key simplification. It separates quantities of utility from the prices of those quantities. The price of U_t is the current unit-expenditure index $\mathcal{P}_t$. The price of U_s for $s > t$ is the expected future unit-expenditure index, brought back to date t through the expected return on saving.

Intertemporal indirect utility

Let

$$A_t \equiv I_t + \sum_{s>t} \frac{\widehat{I}_{s|t}}{\widehat{R}_{s|t}} + k_t \quad (10)$$

denote the present value of current and expected future income plus current wealth, evaluated at date t . Define the intertemporal indirect utility function Z by re-arranging (9) as

$$\begin{aligned} Z(\boldsymbol{\pi}_t, A_t) &= \max_{(U_s)_{s \geq t}} W((U_s)_{s \geq t}) \\ \text{subject to } &\sum_{s \geq t} \pi_{s|t} U_s \leq A_t. \end{aligned} \quad (11)$$

The arguments of Z have the same interpretation as in an ordinary static indirect utility function: they are prices and a budget. The only difference is that the prices are now a whole vector of dated utility prices. If all these dated prices are low, a given amount of resources buys a high intertemporal utility path; if they are high, it buys less.

Assuming homothetic intertemporal preferences, the functional W can be chosen to be homogeneous of degree one. It then follows that the associated indirect utility function Z is homogeneous of degree one in A_t , i.e. for all $\lambda > 0$,

$$Z(\boldsymbol{\pi}_t, \lambda A_t) = \lambda Z(\boldsymbol{\pi}_t, A_t).$$

This homogeneity assumption is not innocuous, but it plays the same role as in the static analysis in Box 1: it makes it meaningful to summarize purchasing power by a single scalar. Doubling all resources, holding the dated price vector fixed, doubles the attainable level of intertemporal utility.

A.2 Purchasing power of current income

We now define the purchasing power of current income. The definition deliberately removes expected future income and inherited wealth from the budget. This does not mean that these resources are unimportant for actual welfare. It means that we are isolating the contribution of the income received at date t , while valuing that income with the same intertemporal price system faced by the consumer.

Definition A.1 (Purchasing power of current income). At date t , the *purchasing power of current income* is defined as

$$\tilde{I}_t \equiv Z(\boldsymbol{\pi}_t, I_t), \quad (12)$$

where $Z(\cdot)$ is the intertemporal indirect utility function defined in (11). Under homogeneity, $\tilde{I}_t$ summarizes the contribution of current income I_t to intertemporal well-being, holding the structure of future prices and returns fixed.

This definition should be read as a decomposition device. Total intertemporal well-being depends on A_t , which includes all current and future resources. The statistic of interest here asks a narrower question: if we take only the income received at date t , and let the consumer

allocate it optimally between current spending and saving, how much intertemporal utility can that income buy? The answer depends on both current prices and the future purchasing power of savings.

A.3 Two stylized expectation regimes

Let

$$\tau_t^C \equiv \frac{\mathcal{P}_t U_t}{I_t}, \quad \tau_t^S \equiv \frac{S_t}{I_t}, \quad S_t \equiv I_t - \mathcal{P}_t U_t. \quad (13)$$

Here τ_t^C is the share of current income spent on current consumption, while τ_t^S is the share saved for future consumption. At the optimum, the saved amount is equal to the present value of the future utility stream financed by current income:

$$S_t = \sum_{s>t} \pi_{s|t} U_s. \quad (14)$$

Hence $\tau_t^C + \tau_t^S = 1$. These shares play the same role as budget shares in a conventional price index. The difference is that they separate current consumption from saving, rather than separating only different categories of current consumption.

In general, there is no closed-form expression for the evolution of $\tilde{I}_t$ under arbitrary expectations. A change observed at date t may affect the whole expected future price and return vector, and therefore the value of saving, in many possible ways. To obtain tractable expressions, we distinguish two stylized cases of how consumers form expectations on future inflation, given observed prices.

A.3.1 Case 1: Fixed expectations on future price levels

The first case captures a current price change that does not revise the expected future purchasing power of savings. Formally, when evaluating the local effect of a change in current income and current prices at date t , the date- t prices of future utility are held fixed:

$$d \log \pi_{s|t} = 0 \quad \text{for all } s > t. \quad (15)$$

Using (3), this can equivalently be written as

$$d \log \left(\frac{\widehat{\mathcal{P}}_{s|t}}{\widehat{R}_{s|t}} \right) = 0 \quad \text{for all } s > t. \quad (16)$$

This corresponds to the case where future price expectations are fixed once and for all. This would be the case for consumers with perfect foresight, or for consumers who do not update their expectations of future prices after temporary deviations from what they had expected.

Under this assumption, the purchasing power of current income satisfies

$$d \log \tilde{I}_t = d \log I_t - \tau_t^C d \log \mathcal{P}_t. \quad (17)$$

Only the part of current income spent immediately is deflated by the current price index. The part that is saved is not deflated by $\mathcal{P}_t$, because its value depends on what it will buy in the future; by assumption, those future dated prices are unchanged.

This is the case in which the distinction between real current income and the purchasing power of current income is most transparent. If all current income were mechanically deflated by $\mathcal{P}_t$, one would implicitly assume that current inflation reduces the real value of saved income in the same proportion as current expenditure. That is not correct when the future purchasing power of savings is held fixed.

A.3.2 Case 2: Fixed expectations on future rates

The second case captures the opposite polar assumption. Consumers observe current prices and current interest rates and adjust their expectations by taking them as the new starting point, while keeping expected future rates of inflation and interest rates fixed. In other words, a higher current price level is expected to raise the whole future price path proportionally, unless expected future inflation rates themselves change.

Formally, assume that the date- t prices of future utility satisfy

$$d \log \pi_{s|t} = d \log \mathcal{P}_t - d \log(1 + r_t) \quad \text{for all } s > t. \quad (18)$$

Equivalently,

$$d \log \left(\frac{\widehat{\mathcal{P}}_{s|t}}{\widehat{R}_{s|t}} \right) = d \log \mathcal{P}_t - d \log(1 + r_t) \quad \text{for all } s > t. \quad (19)$$

A higher current price index raises the expected cost of both current and future utility, because the future price path is re-anchored at today's price level. A higher interest rate r_t , by contrast, makes saving more productive and therefore lowers the current cost of future utility.

Under this assumption, the purchasing power of current income satisfies

$$d \log \tilde{I}_t = d \log I_t - d \log \mathcal{P}_t + \tau_t^S d \log(1 + r_t). \quad (20)$$

Since the expected future price path is indexed to the current price level, current inflation reduces the purchasing power of the whole of current income, not only the part spent immediately. This is why the price term is $-d \log \mathcal{P}_t$, without a consumption-share weight. However, the return on saving affects only the part of income that is carried forward, which is why the interest-rate term is weighted by τ_t^S .

Thus, in this second regime, the standard practice of deflating total current income by the current price index $\mathcal{P}_t$ is appropriate as far as prices are concerned. The additional intertemporal correction comes from changes in the return to saving: a higher return increases the purchasing power of saved income, while a lower return reduces it.

A.4 Including durables

We now add durable goods explicitly, which was the initial motivation of this appendix given the climate-transition applications discussed in the main text. Assume that consumers derive utility from the use of a durable good for which a second-hand market exists. Let u_t^d denote the flow of services from the durable at date t (for instance the kilometers driven), and let k_t^d denote the purchase price of the new durable at date t . Suppose the durable economically

depreciates geometrically at rate δ between t and $t + 1$, such that the good can be resold at price $(1 - \delta)k_t^d$ at date $t + 1$. To simplify, we assume δ is constant over time. Operating costs are not discussed here as they are already included in the period price vector p_t .

A.4.1 Intertemporal budget with durables

When an individual buys a durable in period s , their wealth in $s + 1$ includes the value of the durable net of depreciation. At date t , the consumer hence solves

$$\begin{aligned} \max_{(x_s, u_s^d)_{s \geq t}} \quad & W((U_s)_{s \geq t}) \\ \text{subject to} \quad & U_s = U(x_s, u_s^d) \quad \text{for all } s \geq t, \\ & p_t x_t + k_t^d \mathbf{1}\{u_t^d > 0\} + \sum_{s > t} \frac{\hat{p}_{s|t} x_s + \hat{k}_{s|t}^d \mathbf{1}\{u_s^d > 0\}}{\hat{R}_{s|t}} \\ & \leq I_t + \sum_{s > t} \frac{\hat{I}_{s|t}}{\hat{R}_{s|t}} + k_t + \frac{(1 - \delta)k_t^d \mathbf{1}\{u_t^d > 0\}}{\hat{R}_{t+1|t}} \\ & \quad + \sum_{s > t} \frac{(1 - \delta)\hat{k}_{s|t}^d \mathbf{1}\{u_s^d > 0\}}{\hat{R}_{s+1|t}}. \end{aligned} \tag{21}$$

The final terms in (22) capture the value of durables in future wealth.

By rearranging the sums, one can rewrite the budget constraint as

$$\begin{aligned} \max_{(x_s, u_s^d)_{s \geq t}} \quad & W((U_s)_{s \geq t}) \\ \text{subject to} \quad & U_s = U(x_s, u_s^d) \quad \text{for all } s \geq t, \\ & p_t x_t + c_t^d \mathbf{1}\{u_t^d > 0\} + \sum_{s > t} \frac{\hat{p}_{s|t} x_s + \hat{c}_{s|t}^d \mathbf{1}\{u_s^d > 0\}}{\hat{R}_{s|t}} \leq I_t + \sum_{s > t} \frac{\hat{I}_{s|t}}{\hat{R}_{s|t}} + k_t. \end{aligned} \tag{22}$$

where

$$c_t^d \equiv \frac{r_t + \delta}{1 + r_t} k_t^d \tag{23}$$

and, for $s > t$,

$$\hat{c}_{s|t}^d \equiv \frac{\hat{r}_{s|t} + \delta}{1 + \hat{r}_{s|t}} \hat{k}_{s|t}^d. \tag{24}$$

This is the familiar *user cost* of the durable: the per-period cost of holding one unit of the durable, reflecting both the opportunity cost of funds and depreciation. This user cost formula holds for a new durable good, but it extends to second-hand goods by taking the second-hand market price instead of k_t^d . Relative to (6), the only difference is that the price index at each date now incorporates the user cost of durables through (25) and (26). All of the results from Section A.3 continue to hold once $\mathcal{P}_t$ and $\hat{\mathcal{P}}_{s|t}$ are interpreted as including the user cost of durables. For the main text, this is the relevant conclusion. Intertemporal durable-good effects do not require a separate notion of purchasing power. They require only that the acquisition price of the durable be converted into the appropriate period price

of its service flow. Purchase prices should therefore not be counted entirely at the date of purchase. They should enter the price system in each period in which the durable is held or used, through the user cost formula above. This user cost is the period cost of tying resources up in the durable: it reflects both the opportunity cost of funds and the expected loss of value of the durable over the period.

A.5 Derivation of the two formulas in A.3

This subsection derives (17) and (20). Using the notation introduced above, the date- t problem defining the purchasing power of current income can be written as

$$\begin{aligned} \tilde{I}_t = Z(\boldsymbol{\pi}_t, I_t) &= \max_{(U_s)_{s \geq t}} W((U_s)_{s \geq t}) \\ \text{subject to } &\sum_{s \geq t} \pi_{s|t} U_s \leq I_t. \end{aligned} \quad (27)$$

Recall that $\pi_{t|t} = \mathcal{P}_t$. Let λ_t be the Lagrange multiplier associated with the budget constraint. By the envelope theorem,

$$d\tilde{I}_t = \lambda_t \left(dI_t - \sum_{s \geq t} U_s d\pi_{s|t} \right). \quad (28)$$

Equivalently,

$$d\tilde{I}_t = \lambda_t \left(dI_t - \mathcal{P}_t U_t d \log \mathcal{P}_t - \sum_{s > t} \pi_{s|t} U_s d \log \pi_{s|t} \right). \quad (29)$$

Under the homogeneity assumption, $Z(\boldsymbol{\pi}_t, I_t)$ is homogeneous of degree one in current income. Therefore,

$$\lambda_t = \frac{\partial Z(\boldsymbol{\pi}_t, I_t)}{\partial I_t} = \frac{\tilde{I}_t}{I_t}. \quad (30)$$

Dividing (29) by $\tilde{I}_t$ gives

$$d \log \tilde{I}_t = d \log I_t - \tau_t^C d \log \mathcal{P}_t - \sum_{s > t} \frac{\pi_{s|t} U_s}{I_t} d \log \pi_{s|t}. \quad (31)$$

In Case 1, (15) implies that $d \log \pi_{s|t} = 0$ for all $s > t$. The last term in (31) is therefore zero, and we obtain

$$d \log \tilde{I}_t = d \log I_t - \tau_t^C d \log \mathcal{P}_t, \quad (32)$$

which is (17).

In Case 2, (18) implies that all future dated utility prices move by the same proportional amount:

$$d \log \pi_{s|t} = d \log \mathcal{P}_t - d \log(1 + r_t) \quad \text{for all } s > t.$$

Hence

$$\begin{aligned} \sum_{s > t} \frac{\pi_{s|t} U_s}{I_t} d \log \pi_{s|t} &= \left(\sum_{s > t} \frac{\pi_{s|t} U_s}{I_t} \right) (d \log \mathcal{P}_t - d \log(1 + r_t)) \\ &= \tau_t^S (d \log \mathcal{P}_t - d \log(1 + r_t)), \end{aligned} \quad (33)$$

where the second equality uses (14). Substituting this expression into (31) gives

$$\begin{aligned}
d \log \tilde{I}_t &= d \log I_t - \tau_t^C d \log \mathcal{P}_t - \tau_t^S (d \log \mathcal{P}_t - d \log(1 + r_t)) \\
&= d \log I_t - (\tau_t^C + \tau_t^S) d \log \mathcal{P}_t + \tau_t^S d \log(1 + r_t) \\
&= d \log I_t - d \log \mathcal{P}_t + \tau_t^S d \log(1 + r_t),
\end{aligned} \tag{34}$$

which is (20).